High Contrast Interferometric Second Harmonic Generation Microscopy Probes the Polarity of Fibrils in Complex Collagenous Tissues

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Abstract: We implemented for the first time Interferometric SHG microscopy, using femtosecond pulses, to image the relative fibrils polarity in cartilage and show that SHG intensity depends on the local ratio of fibrils with opposite polarities.
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1. Introduction

In recent years, Second Harmonic Generation (SHG) microscopy has emerged as a powerful technique to image *in situ* non-centrosymmetric structures in tissues. In particular, SHG appears to be a very sensitive structural probe of the 3D architecture of fibrillar collagen in biological tissues [1-8].

Collagen is a major structural protein of vertebrates and the principal constituent of the extracellular matrix. There exist twenty eight types of collagen, all being built from the same elementary macro-molecule, namely the tropocollagen, formed as an assembly of polypeptide chains arranged in a triple helix. Among them, fibrillar collagens (such as type I and II) result from the staggered alignment of triple helices to form fibrils with 10 to 300 nm of diameter and a few microns length [9-11]. At the macroscopic scale, collagen fibrils are arranged in a 3D meshwork with architecture specific from one tissue to another, in order to comply with the requirements imposed by the physiological functions of tissues.

SHG microscopy is a coherent imaging technique characterized by an intrinsic optical sectioning, a high penetration depth in scattering tissues and reduced phototoxicity and photobleaching. In fibrillar collagen, because of the tight alignment of harmonophores within fibrils, SHG is coherently amplified resulting in strong signals and highly contrasted images. Extensive literature is available about the molecular and structural origin of SHG signal [12-16] and its relation with the overall collagen organization as revealed, for example, by Polarization-resolved SHG (P-SHG) [17-19].

However, due to the coherence of SHG, the signal intensity depends not only on the density and the overall organization of harmonophores, but also on the relative polarity of the collagen fibrils within the focal volume. In SHG imaging, the phase of the signal, which contains additional structural information, is lost in the measurements. This has impeded the study of the nanoscale arrangement of the fibrils' polarity.

Recently, Interferometric SHG (I-SHG) microscopy has been proposed to overcome this limitation by measuring the phase of the signals. In I-SHG microscopy, second harmonic is generated twice: first outside the microscope, to provide a reference SHG beam, and secondly within the sample, to probe the fibrils' polarity. By varying the phase relationship between the two second harmonic signals that are interfering, the relative phase of the second-order susceptibility, or equivalently the relative orientation of the second harmonic emitters, can be retrieved pixel by pixel. I-SHG has been first applied to the study of myosin filaments in muscles [20] and highly organized collagen bundles in tendons [21]. However, the investigation of tissues exhibiting more complex collagen architecture, such as articular cartilage, has been hindered so far by the low interferometric contrast obtained with picosecond pulses laser.

In this work we implemented for the first time I-SHG using femtosecond pulses and applied it to study the collagen meshwork in cartilage. To do so, we developed a collinear setup to compensate for the dispersion introduced in the microscope as to optimize both spatial and temporal overlap. The results are analyzed in regards of numerical simulations showing that the phase measurements can be related to the local ratio of fibrils with opposite polarities, which is responsible for the strong variations of signal intensity in our images. A comparison with standard and polarization-resolved SHG highlights the role of both tissue organization and relative fibril polarity in determining the SHG signal intensity. This work illustrates how the complex nanoscale organization of harmonophores within collagen fibrils influence the coherent building of non-linear signals within the focal volume.

2. Methods

SHG imaging is performed using a custom-built sample scanning microscope (figure 1.a). For I-SHG microscopy, two things need to be added to the setup: a reference quartz and a glass window [20,21]. The reference Y-cut quartz plate allows us to generate a reference SHG beam with a known phase. Then a second SHG beam is generated in the sample at the focus of the objective. These two SHG beams (reference and sample SHG) interfere on the detector to generate the final signal in every pixel, given by

$$I(\phi_{ref}) = I_{ref} + I_{exp} + 2\sqrt{I_{ref} I_{exp}} \cos(\phi_{exp} - \phi_{ref}) \qquad (1)$$

where I_{ref} and I_{exp} are respectively the SHG intensity in the reference quartz and in the sample. The reference phase (ϕ_{ref}) is adjusted by rotating the glass window. We then isolate the interferometric term by subtracting, pixel by pixel, two images taken with a π phase shift:

$$I(\phi_{ref}) - I(\phi_{ref} + \pi) = 4\sqrt{I_{ref} I_{exp}} \cos(\phi_{exp} - \phi_{ref}) \qquad (2)$$

This allow us to extract the relative phase of the sample SHG (ϕ_{exp}) in every pixel.

In order to optimize the imaging contrast, we use here a femtosecond laser to generate second harmonic in the sample. However, in this case, a fine control of both spatial and temporal overlap between reference and sample SHG beams is required to obtain interferences on the detector. Hence, in the setup we developed, both reference SHG and fundamental beams are collinear which ensures both spatial overlap and phase stability of the system. The temporal overlap is then optimized using two calcite wedges that compensate for the dispersion introduced by all the optics, mainly the objective.

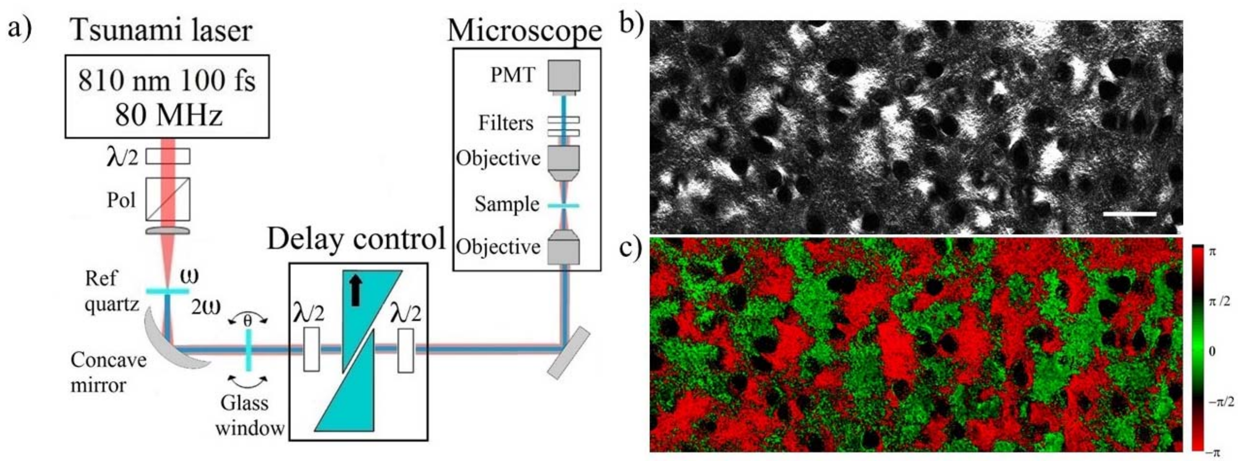

Figure 1: a) Experimental setup used for I-SHG microscopy. The delay control is achieved through two calcite prisms placed between two half-waveplates at 810 nm and full waveplates at 405 nm so that the fundamental beam exhibits a retardation compared to the reference SHG beam. The black arrow indicates the optical axis. (b) SHG image and (c) I-SHG image of articular cartilage. Pixels with SHG intensity below 50 counts in a) are displayed in black in the I-SHG image. Scale bar: 100 µm.

3. Results

In the SHG image of articular cartilage (figure 1.b) the black circles correspond to holes left by the cells (chondrocytes) after the fixation. We can immediately distinguish two types of region: a large part of the image exhibits a speckle-like aspect with low average intensity whereas several areas present high and almost uniform SHG signals. The general low intensity aspect of the image is expected from the cartilage micro-structure composed of a random-like collagen II meshwork. In contrast, the high intensity domains, mainly located around the cells, indicate regions with highly organized collagen distribution.

Figure 1.c) shows the I-SHG image of the same region of interest in the sample and presents, pixel by pixel, the phase of the SHG signals measured. Interestingly, the I-SHG image reveals large regions with a constant phase, which correspond to the high intensity domains in figure 1.b). On the contrary, the speckle-like domains in the SHG images present a quasi-random phase in the I-SHG image.

As SHG is a coherent process, the two predominant phases can be related to opposite fibrils' polarity. Indeed, individually, two fibrils with opposite polarities will generate π phase-shifted SHG signal, with the same intensity. However, as in every pixel SHG results from the contribution of several fibrils within the focal volume, the relative polarity between adjacent fibrils can highly affect the signal intensity. By further investigation we find, in regards with numerical simulations, that the phase distribution in the I-SHG images can be related to the local area fraction of fibrils with opposite polarities. A ratio close to 0.5 leads to a quasi-random phase and speckle-like SHG intensity region. On the contrary, a ratio close to 0 or 1 indicates a highly polarized domain and therefore a well-defined phase which results in a high SHG intensity.

4. Conclusion

I-SHG, by measuring the phase of SHG signals, allows to probe the relative orientation of collagen fibrils in the focal volume and therefore provides additional information both on the samples' structure and the SHG process in tissues. In particular, this work highlights the impact of the relative fibrils polarity on the coherent building of SHG signals, which is particularly crucial in biological tissues presenting a complex collagen architecture such as articular cartilage.

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